Applications and Instructions for the AILEIKE ALKW851D Zero-Degree Thermostat

The ALKW851D zero-degree thermostat is a high-precision constant temperature device that replaces traditional ice-water mixtures, providing a stable output of a standard temperature field at 0℃. Equipped with multiple temperature measurement jacks, it adopts a semiconductor cooling and silent heat dissipation design, ensuring stable operation without drift. It serves as a fundamental supporting device for thermal calibration and precision temperature measurement. The device is locked at a constant temperature of 0℃ by default, eliminating the need for manual temperature adjustment. It is easy to operate and can work continuously and stably for extended periods.

1. Technical indicators

name	Zero-degree thermostat	Measure the diameter of the hole	Φ9mm
Model number	ALKW851D	Number of holes measured	6 holes
precision	0℃±0.05℃	Power	200W
display resolution	0.01℃	Power supply	Single-phase AC 220V
stability	±0.005℃	Overall dimensions	150×260×370mm
Working environment temperature	22℃±3℃	weight	7.2kg

II. Usage

1. Connect to the power supply, which is a single-phase AC 220V.

2. Press the power switch, and if the constant temperature indicator light is on, it indicates that the power supply of the semiconductor cooler has been connected and the semiconductor cooler is already working. The temperature in the working area of the thermostat starts to decrease. At this time, the temperature controller displays the temperature of the thermostat, and there is air discharged from its upper part. The presence of air cooling indicates that the fan is working properly.

3. When the temperature in the working area of the zero-degree thermostat drops to 0 degrees, the thermostat indicator light starts to flicker. At this point, the zero-degree thermostat has entered a constant temperature state. After entering the constant temperature state for about 15 minutes, the zero-degree thermostat can be

used as a reference junction temperature for thermocouples, enabling temperature measurement, verification, calibration, and other related tasks.

4. After all tasks are completed, first remove all thermocouple probes one by one and confirm that there are no objects left in the temperature measurement jacks. Turn off the equipment power switch and disconnect the external power supply.

III. Calibration method

1. Test conditions: room temperature of 22℃±3℃, second-class standard platinum resistance thermometer, and 0.005-level six-digit digital multimeter.

2. Calibration method: Insert the second-class standard platinum resistance thermometer into the socket of the zero-degree thermostat and connect it to the testing system. Turn on the power switch and take measurements after 40 minutes. Measure the deviation between the actual temperature of the zero-degree thermostat and zero degrees, and observe its stability.

3. When the temperature reaches the calibration temperature, press the ↻ key located at the leftmost end of the lower part of the temperature control meter. After a few seconds, a menu will pop up. Continue pressing the ↻ keys. When the "SC" is displayed in the PV window, you can use the ◄ , ▲, and ▼ keys to adjust the value in the SV window. Measure the actual temperature of the zero-degree thermostat with a standard platinum resistance thermometer. When the displayed value on the temperature control meter is higher than this temperature, increase the "SC" value by the difference between the two, and vice versa.

IV. Applicable Scenarios

1. Metrological verification of thermocouples

The thermocouple verification procedure can be used for the indexing verification and accuracy calibration of various types of base metal and precious metal thermocouples, such as K, T, J, S, R, and B.

2. Precise temperature measurement test in the laboratory

It is suitable for various temperature calibration and temperature measurement experiments in scientific research laboratories and quality inspection and testing laboratories, providing a standardized 0℃ constant temperature reference for the accuracy calibration of temperature sensors, temperature measurement instruments, and acquisition modules.

3. Thermal calibration on industrial site

It is widely used in industrial sites across various sectors such as electricity, chemical industry, metallurgy, and manufacturing.

4. Teaching and practical training in colleges and universities

Used for practical teaching of thermal engineering courses in colleges and universities, it visually demonstrates the principle of thermocouple temperature measurement and cold junction compensation technology.





power industry



Metallurgical industry



chemical industry

V. Precautions for Use

1. It is strictly prohibited to pour liquids, debris, and metal scraps into the temperature measurement jack to prevent internal components from being affected by moisture, short-circuited, or blocked, which may cause damage to the refrigeration module.

2. It is prohibited to forcefully insert probes of mismatched sizes to avoid squeezing and damaging the internal temperature equalizing block, which could affect the accuracy of the temperature field.

3. It is strictly prohibited to block the cooling vents during equipment operation, as this can lead to poor heat dissipation, ineffective cooling, overheating of the equipment, temperature drift, and equipment malfunction.

4. The internal temperature control parameters of the equipment have been factory calibrated and locked. Non-technical personnel are strictly prohibited from dismantling the equipment or modifying the PID temperature control parameters. Unauthorized changes may lead to loss of accuracy and disqualification from warranty.

5. The equipment can operate continuously for a long period of time. When it is left idle for an extended period, it is recommended to power it on and run it for 30 minutes every month to prevent moisture and maintain the internal components, thereby extending the service life of the equipment.

6. The semiconductor cooler is a ceramic device. Please avoid strong vibration and impact.

VI. Conclusion

The ALKW851D zero-point thermostat is a high-precision thermocouple 0℃ reference device. Compared to traditional ice-water mixtures, it features high accuracy, good stability, and no impurity contamination. The device maintains a constant temperature of 0℃ and automatically cools down upon startup. Once stabilized, it can be used for various thermocouple calibrations, precise temperature measurements in laboratories, on-site industrial calibrations, and teaching and training scenarios. When in use, simply place it properly, ensure

proper heat dissipation, and maintain sufficient temperature stability for stable operation. It is strictly prohibited to block the air vents or contaminate the jacks on a daily basis. Regular power-on maintenance can maintain measurement accuracy over the long term and meet the requirements of thermal engineering testing and value traceability.